

User Interface

1. General Interfaces (All Users)

Interface Name	Use Case Supported	Description
Login/Authentication Page	User Authentication	Allows all users (Admin, Teacher, Student, Parent) to enter a unique username and password to log in and verifies credentials ² .
Main Dashboard	Announcements and Communication	The landing page that displays role-specific quick links and announcements ³³³ .
User Profile Management	User Management (Edit Profile)	Allows users (Teachers) to view/update their personal details, excluding roles or permissions ⁴ .
Search and Filter Page	Reports and Data Management	Provides options to quickly find and filter records (e.g., past announcements, student/teacher records) ⁵⁵⁵⁵ .
Password Reset/Recovery	System Management & Security	Allows users to reset their forgotten passwords ⁶ .

2. Admin Interfaces

Interface Name	Use Case Supported	Description
Manage User Accounts Form	Manage User Accounts, Add/Edit Student Info, Teacher Management	Allows the Admin to add, view, edit, or remove accounts for Students and Teachers ⁷⁷⁷⁷⁷⁷ .
Teacher-Class Assignment	Assign Teachers to Classes	Interface to assign teachers to specific classes or subjects ⁸ .
Schedule Finalization Screen	Publish Schedule	Interface to review the final draft of the schedule and finalize/publish the master schedule ⁹⁹⁹⁹ .
Report Generation Tool	Generate Reports	Allows the Admin to select parameters (dates, classes, etc.) to generate grade and attendance reports ¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰ .

3. Teacher Interfaces

Interface Name	Use Case Supported	Description
Attendance Recording Form	Record Attendance	A list-based or visual interface for marking students present or absent for a class session ¹¹¹¹¹¹¹¹¹¹ .
Grade Entry/Update Form	Enter or Update Grades	Allows the Teacher to enter and update student grades for assignments and subjects, with auto-calculation options ¹² .

Interface Name	Use Case Supported	Description
Study Material Upload	Upload Study Materials	A form to upload files (Study Materials) and link them to specific classes/subjects ¹³ .
Class Announcement Composer	Announcements and Communication	Allows the Teacher to write announcements, select recipients, and choose notification delivery methods (in-system, email, SMS) ¹⁴¹⁴¹⁴¹⁴¹⁴¹⁴¹⁴¹⁴¹⁴¹⁴¹⁴¹⁴ .

4. Student & Parent Interfaces

Interface Name	Use Case Supported	Description
Student Grades View	View Grades	Displays the student's grades, organized by course, due date, or category ¹⁵¹⁵¹⁵¹⁵ . This view is also accessible to the Parent ¹⁶ .
Attendance History View	View Attendance	Displays the student's personal attendance history and records ¹⁷¹⁷¹⁷¹⁷ .
Schedule Viewing Page	Publish Schedule (View)	Displays the student's class schedule in a daily/weekly list or interactive calendar view ¹⁸¹⁸¹⁸¹⁸ .
Announcement Feed	Announcements and Communication	A dedicated area (feed or dashboard banner) for reading school and class-related announcements ¹⁹ .

System Interface

1. External System Interfaces (APIs and Gateways)

These interfaces facilitate communication with components outside the immediate system boundary, primarily for notifications.

Interface Name	Type	Source → Destination	Short Description
SMS Notification API	External API (Web Service)	SMS -> External SMS Gateway	Sends urgent text messages/SMS notifications to recipients for urgent announcements.
Email Transport Service	SMTP / External API	SMS -> External Email Server	Handles the transmission of non-urgent and urgent email notifications (e.g., parent notifications, password resets).
Database Backup Interface	Automated Feed (Data Stream)	SMS -> External Storage (e.g., Cloud Backup)	Periodically copies the central database contents for secure data archiving and disaster recovery.

2. Internal Sub-System Interfaces (Modules and Database)

These interfaces manage the flow of data between the application logic modules and the central database, which acts as the core repository for all system information.

Interface Name	Type	Source → Destination	Short Description
User Data CRUD Interface	Database (SQL Operations)	System Modules <-> Central Database	Standard operations (Create, Read, Update, Delete) for managing all user account and profile data (Admin, Teacher, Student, Parent).
Transactional Logging Interface	Database (Append-only)	System Modules -> Central Database	Records and stores all system activities for audit purposes (e.g., logins, data modifications, announcements sent).
Conflict Check Service	Inter-Module Call	Scheduling Module->Internal Logic	Sends proposed scheduling details to be validated against existing room/teacher assignments to ensure no double-booking occurs.
Schedule Data Exchange	Database (Read/Write)	Scheduling Module <-> Central Database	Writes and retrieves current or draft class schedule data, including teacher/room assignments.
Grade Calculation Logic	Inter-Module Call	Grade Entry Module -> Internal Logic	Sends raw scores and weighting rules to calculate and retrieve finalized student averages or overall subject grades.

